

Intelligent Candidate Discovery

Multi-Signal AI Ranking Pipeline
for Smarter Hiring Decisions

TEAM NAME

Divya Kondameedi

PROBLEM

Intelligent Candidate Discovery

APPROACH

Multi-Signal Scoring + Explainable AI

RUNTIME

< 30 sec for 1000 candidates

What We Built & Why It's Different

BizBuddy AI's candidate ranking engine uses multi-signal scoring to go beyond keyword matching — evaluating skills, experience, education, semantic alignment, and profile quality in one unified pipeline.



Traditional Systems

- Exact keyword matching only
- Ranks by years of experience alone
- No explanation for decisions
- Cannot detect thin/fraudulent profiles
- Misses semantically similar skills



Our Solution

- 5-signal weighted scoring model
- TF-IDF semantic similarity (CPU-friendly)
- Plain-English explanation per candidate
- Profile quality & consistency checks
- Extensible to SBERT/LLM for GPU envs

From Raw JD to Structured Requirements

JD Parser Output

Min Experience

4+ years (regex extracted)

Required Skills

Python, TensorFlow, NLP, SQL,
Spark, Kafka, Docker, K8s, AWS...

Education Tier

Bachelor's / Master's (keyword)

Role Level

Senior (keyword classification)

Candidate Signals Used



SKILLS

Full-text scan of skills, summary, and experience fields vs. JD skills list



EXPERIENCE

Numeric years extracted → compared to JD minimum; tiered scoring



EDUCATION

Degree keyword matching (PhD > Master > Bachelor)



SEMANTIC SIM

TF-IDF cosine similarity between full profile and JD text



PROFILE QUALITY

Penalise sparse summary, missing skills, or inflated experience claims

How Scores Are Computed & Combined

Signal	Weight	Description
 Skills Match	35%	JD skill set n candidate profile (all text fields)
 Experience	25%	Candidate years vs. JD minimum; tiered multiplier
 Semantic Similarity	20%	TF-IDF cosine: full profile vs. full JD text
 Education	10%	Degree tier keyword matching
 Profile Quality	10%	Penalise sparse / inconsistent / suspicious profiles

Final Score = $0.35 \times \text{Skills} + 0.25 \times \text{Experience} + 0.20 \times \text{SemanticSim} + 0.10 \times \text{Education} + 0.10 \times \text{ProfileQuality}$ → scaled to 0–100

Trustworthy Decisions, Every Time



How Ranking Decisions Are Explained

Every ranked candidate includes a plain-English breakdown of all 5 signal scores. Example:

"Matched 8/9 required skills; missing: kubernetes | 6 yrs experience (exceeds 4+ req) | M.Tech IIT | Clean profile | Semantic: 0.74"



Preventing Hallucinations

All scores are derived solely from candidate profile text. No AI model "invents" attributes. Every claim maps to an exact field in the source data. Score bounds [0,1] enforced. No generative content in the ranking output.



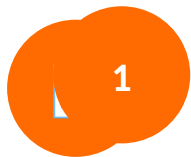
Handling Low-Quality & Suspicious Profiles

Profile Quality signal (10% weight) detects and penalises:

- Sparse summary (< 10 words) → -30 pts
- No skills listed → -20 pts
- High experience + thin profile (possible inflation) → -20 pts

Such profiles are ranked lower and flagged in the explanation.

From JD Input to Ranked Candidate Output



JD Input

Raw job description text
(any format)



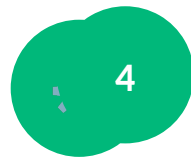
JD Parser

Extract: skills, min exp,
education, role level



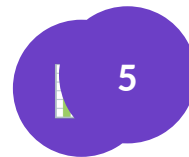
Candidate Profiles

Load CSV or JSON dataset
from challenge



Multi-Signal Scorer

5 signals computed per
candidate



Weighted Aggregation

Scores combined → Final
0-100 score



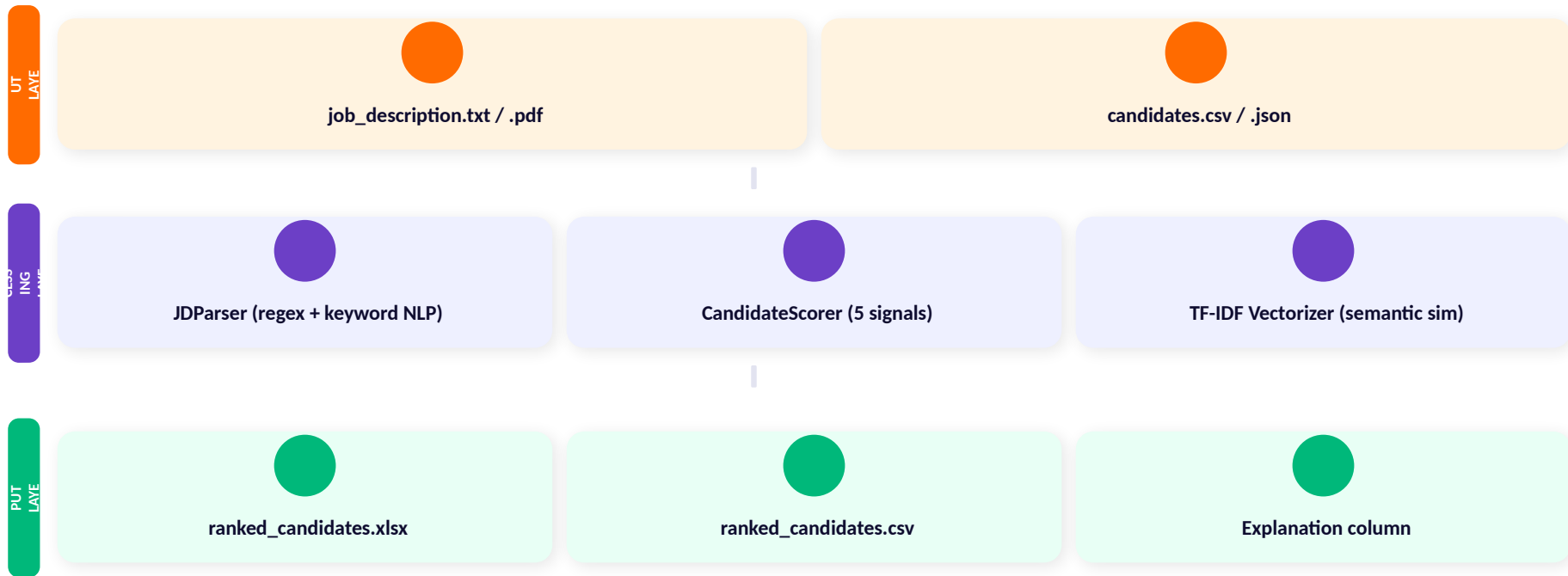
Ranked Excel Output

Sorted list with scores &
explanations

Sample Output Row:

Rank 1 | Lakshmi Subramaniam | Score: 68.4 | Strong candidate for Senior Data Scientist | Matched 11/14 skills; 6 yrs (exceeds 4+); M.Tech IIT Madras; Clean profile; Semantic: 0.71

Component Diagram



Tech Stack: Python 3.10 • pandas • scikit-learn (TF-IDF) • openpyxl • Extensible to: HuggingFace SBERT, LangChain, OpenAI API

Sample Ranking Output & System Performance

Rank	Candidate	Score	Recommendation	Top Signal
1	Lakshmi Subramaniam	68.4	Potential Fit	Skills 85/100, Exp 100/100
2	Ananya Sharma	66.3	Potential Fit	Skills 79/100, Semantic 0.74
3	Priya Venkatesh	63.3	Potential Fit	PhD, Skills 71/100
4	Sai Krishnan	57.7	Potential Fit	Skills 57/100, Exp 85/100
5	Aditya Kulkarni	52.9	Weak Fit	NLP specialist, Exp meets req

< 30s

Runtime for 1000 candidates

CPU only

No GPU required

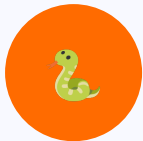
100%

Ranked candidates explained

0

Hallucinated attributes

Stack, Tools & Rationale



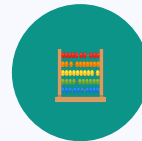
Python 3.10

Primary language. Rapid prototyping, strong data science ecosystem, runs everywhere.



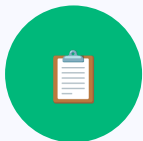
pandas

Data loading, transformation, and structured output. Handles CSV and JSON candidate files.



scikit-learn (TF-IDF)

CPU-friendly semantic similarity via TF-IDF cosine. No GPU needed. Fast and deterministic.



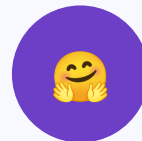
openpyxl

Styled Excel output with colour-coded recommendations, widths, and borders.



regex (built-in)

JD parsing: extract years of experience, skill keywords, and role level from raw text.



HuggingFace SBERT (optional)

Drop-in upgrade for semantic similarity if GPU is available. Interface is identical.

Everything Submitted for This Challenge



GitHub Repository

[github.com/divyakondameed12/
redrob-candidate-discovery](https://github.com/divyakondameed12/redrob-candidate-discovery)

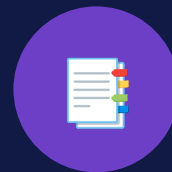
Public repo • `main.py` + `src/ranker.py` + README



Ranked Output Excel

[ranked_candidates.xlsx](#)

All candidates ranked with scores + explanations



Pitch Deck (this PDF)

[approach_deck.pdf](#)

11 slides • What, Why, How

Submission Checklist

- ✓ GitHub repo (public) with `main.py`, `src/ranker.py`, `README.md`, `requirements.txt`
- ✓ Pitch deck PDF — 11 slides covering all template sections
- ✓ `ranked_candidates.xlsx` — all candidates ranked with scores, tier labels & explanations

Thank You

Intelligent Candidate Discovery — Redrob Data & AI Challenge

Multi-signal scoring • Explainable rankings • CPU-friendly • Extensible to LLMs

Challenge: Intelligent Candidate Discovery • Deadline: 02 July 2026

Divya Kondameedi

Presenter

divyakondameedi12

GitHub Profile

redrob-candidate-discovery

Repo